Optimizing Sales, Inventory and Margin For Eesbee Garments

A Mid-Term report for the BDM capstone Project

Submitted by

Name: Pranathi Ayyadevara

IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

Table of Contents

1. **Executive summary** 2
2. **Proof of originality of the Data** 2
3. **Metadata and Des criptive Statistics** 3
   - 3.1 Metadata 3
   - 3.2 Descriptive Statistics 5
4. **Analysis Process, Tools and Methods** 7
   - 4.1 Process Flow 7
   - 4.2 Tools 8
   - 4.3 Analytical Methods by Domain 9
5. **Results and Findings** 10
   - 5.1 Margin Diagnostics 10
   - 5.2 Inventory Analysis 11

1. Executive summary

Eesbee is a B2C single-door garment store located in Alkapuri, Hyderabad and sells variety of branded items. The project was initiated to build a reliable and scalable diagnostic framework for retail inventory analysis. This mid-term review presents a preliminary diagnostic built on data collected from the store POS system. Originality of the dataset was established through store visits, video documentation, and system-level validation. The Metadata structures were defined to support SKU-level granularity, enabling reproducible diagnostics across brands, categories, and time periods. Descriptive statistics and early visualizations were used to surface patterns in inventory movement, pricing behavior, and reconciliation gaps.

A *Python Framework* is being developed for analysis and is progressing well. The application is built with emphasis placed on modularity, traceability, and business relevance. At this stage, only a limited set of graphs and diagnostics has been included to illustrate key trends. Methods such as Margin Diagnostics, Inventory Analysis, deadstock identification, discrepancy tracking, and distribution analysis were applied to highlight systemic inefficiencies and margin risks.

Preliminary findings to date suggest that the issues observed are structural rather than isolated, with recurring patterns in inventory mismatches, poor sell-through, and return-heavy price bands. These insights remain directional and may change as we move to next steps. The full analytical framework, along with expanded visualizations, refined logic, and SKU-level recommendations, will be presented in the *Final Report*, when the codebase has been stabilized and all diagnostics have been completed. A project folder on google driver is created to store all data files, generated result files, documents, original plot images and source code.

2. Proof of originality of the Data

Eesbee Garments is owned by two partners and has eight employees. The majority of the project related communication was conducted with one of the partners, both in-store and over the phone. The details of the store, along with proof of data originality, are provided in the Table **MTT-01** below.

Proof of Originality			
Store Name	EESBEE Garments	Contact	+91-9014885517, (040) - 42607224
Address	H.N-11-13-376, Road No: 4, Alkapuri, Hyderabad-35	GST Number	36AADFE3960A2ZU

File Details			
Sl. No.	**Name**	**Description**	**Link**
1	Field Log	Details of in-person field visits and online interviews	documents\Field log.pdf
2	Store Artifacts	Sample purchase invoice, sale bill, expense tracking chart	documents\Artifacts.pdf
3	Business Letter	Letter from business for project engagement and data sharing	documents\Business Letter.pdf
4	Photographs	Photographs of the store and with owner/staff	documents\Photographs.pdf
5	Video	Video of store introduction, owner's interaction and data discussion	videos\Video.mp4
6	Data Folder	Primary data folder for input CSV files, output result files	data\input, data\output
7	Project Source	Python project source code and related files	source
8	Plots/Graphs	Full size images of Plots and Graphs	images

Table **MTT-01**: Proof of Originality Details

NOTE: All files are stored in the ***BDM Project Sep 2025*** folder on the google drive (link: https://drive.google.com/drive/folders/1wg_f42GfTeMN3rEyTI3y-qcsgQIa793S?usp=sharing)

3. Metadata and Descriptive Statistics

Eesbee POS system offers basic, yet valuable reports that can be exported in CSV format. The metadata section provides the overview of data, its structure, and the purpose for useful columns. The basic statistics for all input data is provided in the descriptive statistics section.

3.1 Metadata

Opening Stock and Closing Stock: The opening stock as of April 1st 2023 (*opening_stock.csv*) contains **6295** records and the closing stock as of September 30th 2025 (*closing_stock.csv*) contains **8767** records. Both files have **14** columns each, out of which **7** columns are used for the analysis and are described in the Table **MTT-02** below. These columns form the foundation for inventory valuation and margin diagnostics in the subsequent sections.

Opening & Closing Stock Column Details			
Sl. No.	**Column Name**	**Data Type**	**Description**
1	Stock No	String	Unique identifier for the item (SKU). Majority are numeric, but some have letters.
2	Product	String	Name of the product, for example, SHORTS
3	Brand	String	Name of the brand, for example, Jockey
4	Cost Price	Float	Cost price for the item, generally referred to as COGS. The unit value is INR (₹)
5	Retail Price	Float	The price marked on the product by the manufacturer, also refereed as MRP. The unit value is INR (₹)
6	Closing Bal.Qty	Integer	Quantity of items available in store as of the given date
7	Closing Bal.Val	Float	The total price paid for the set of items (Cost Price * Closing Bal.Qty). The unit value is INR (₹)

Table **MTT-02**: Metadata for Closing and Opening Stock

Purchase Data: The purchase data between April 1st 2023 and September 30th 2025 (*purchase_data.csv*) contains **48338** records. The file has **17** columns out of which **9** columns are used and are described in the Table **MTT-03** below. These columns form the foundation for inventory valuation and margin diagnostics in the subsequent sections.

Purchase Data Column Details			
Sl. No.	**Column Name**	**Data Type**	**Description**
1	Voucher Date	Datetime	Date and time of the item purchase
2	Voucher No	String	Identification number for the purchase, typically in P/### format
3	Particulars	String	Name of distributor supplying the items
4	Stock No	String	Unique identifier for the item (SKU). Majority are numeric, but some have letters
5	Product	String	Name of the product, for example, SHORTS
6	Brand	String	Name of the brand, for example, Jockey
7	Retail Price	Float	The price marked on the product by the manufacturer, also refereed as MRP. The unit value is INR (₹)
8	Purchase Qty	Integer	Quantity of items purchased by the Eesbee
9	Net Amount	Float	Amount paid to the distributor. The unit value is INR (₹)

Table **MTT-03**: Metadata for Purchase Data

Purchase Return Data: The purchase return data between April 1st 2023 and September 30th 2025 (*purchase_return_data.csv*) contains **10803** records. Return of the stock could happen because of

various reasons like unsold, consignment or damaged items. The file has **16** columns out of which **9** columns are used and are described in the Table **MTT-04** below.

Purchase Return Data Column Details			
Sl. No.	**Column Name**	**Data Type**	**Description**
1	Voucher Date	Datetime	Date and time of the item purchase
2	Voucher No	String	Identification number for the purchase, typically in PR/### format
3	Particulars	String	Name of distributor supplying the items
4	Stock No	String	Unique identifier for the item (SKU). Majority are numeric, but some have letters
5	Product	String	Name of the product, for example, SHORTS
6	Brand	String	Name of the brand, for example, Jockey
7	Retail Price	Float	The price marked on the product by manufacturer, also refereed as MRP. The unit value is INR (₹)
8	Purch. Return Qty	Integer	Quantity of items returned to the distributor by the Eesbee
9	Net Amount	Float	Net Amount value of the return. The unit value is INR (₹)

Table **MTT-04**: Metadata for Purchase Return Data

Stock Verification Data: The stock verification/adjustment data between April 1st 2023 and September 30th 2025 (*stock_verification_data.csv*) contains **7034** records. The stock verification are adjustments are made to the inventory after physical verification. The adjustments could arise for various reason, like incorrect receiving, theft or incorrect checkout quantity. The file has **18** columns of which **10** columns are used and are described in the Table **MTT-05** below.

Stock Verification Data Column Details			
Sl. No.	**Column Name**	**Data Type**	**Description**
1	Voucher Date	Datetime	Date and time of the adjustment after physical stock verification
2	Voucher No	String	Identification number for the physical stock verification, typically in PS/### format
3	Stock No	String	Unique identifier for the item (SKU). Majority are numeric, but some have letters
4	Product	String	Name of the product, for example, SHORTS
5	Brand	String	Name of the brand, for example, Jockey
6	Retail Price	Float	The price marked on the product by manufacturer, MRP. The unit value is INR (₹)
7	Book Stock	Integer	Quantity of item per POS inventory system
8	Trans Qty. (Phy Stk Qty)	Integer	Actual physical quantity in the store
9	Difference Qty	Integer	Adjustment quantity to make system and physical stock match
10	Physical Stock Value	Float	Amount of physical stock verified. The unit value is INR (₹)

Table **MTT-05**: Metadata for Stock Verification Data

Sales Data: The sales data (*sales_data.csv*) contains **93777** records, of which **91486** are sale records and **2291** are customer returned records. Each item in a customer bill is recorded as a separate entry, allowing for detailed line-level analysis. The file has **20** columns out of which **9** columns are used for the analysis and are described in the Table **MTT-06** below.

Sales Data Column Details			
Sl. No.	**Column Name**	**Data Type**	**Description**
1	Voucher No	String	Identifier for sale transaction, typically in S/### format
2	Voucher Date	Datetime	Date and time of the item sale
3	Stock No	String	Unique identifier for the item (SKU). Majority are numeric, but some have letters
4	Product	String	Name of the product, for example, SHORTS
5	Brand	String	Name of the brand, for example, Jockey
6	Sales Qty	Integer	Quantity of items sold by the Eesbee
7	Bill Level Disc. Amt.	Float	The total discount given to customer. The unit value is INR (₹)
8	Net Amount	Float	The final amount paid by the customer after discount. The unit value is INR (₹)
9	Retail Price	Float	The price marked on the product by the manufacturer, MRP. The unit value is INR (₹)

Table **MTT-06**: Metadata for sales data

Expense Data: The aggregated overhead expense between April 1st 2023 and September 30th 2025 (*expense_data.csv*) contains 3**0** records, with one record per month. This data file has been created manually based on the information extracted from the Tally system. It contains **2** columns and are described in the Table **MTT-07** below.

Expense Data Column Details			
Sl. No.	**Column Name**	**Data Type**	**Description**
1	Month	Datetime	Month and year for the expense record
2	Expense	Float	The aggregated overhead expense for the month. The unit value is INR (₹)

Table **MTT-07**: Metadata for Expense Data

3.2 Descriptive Statistics

To identify patterns, central tendencies and the spread of the data, calculations like mean, median, standard deviation and IQR are calculated on all input data sets. These statistics are calculated using Python Pandas library *dataframe.describe()* and *dataframe.sum()* methods are transposed using *dataframe.transpose()* method.

Opening Stock and Closing Stock: Descriptive statistics for Opening and Closing stock are provided in the Table **MTT-08** below.

Dataset	Metric	Count	Sum	Mean	St.Dev.	Min	25%	50%	75%	Max
Opening Stock	Cost Price	6295	2832744.395	449.9991	241.7094	0	280	395.81	599.33	1821
Opening Stock	Retail Price	6295	4329414	687.7544	384.0127	50	416	590	895	3999
Opening Stock	Closing Bal.Qty	6295	13984	2.221446	3.220557	1	1	1	2	110
Opening Stock	Closing Bal.Val	6295	5227318.57	830.3921	1013.341	0	366	590.18	881	28336
Closing Stock	Cost Price	8767	4039375.78	460.7478	258.6495	0.03	270	407.78	602.95	2995
Closing Stock	Retail Price	8767	6224650	710.0091	410.684	35	415	599	919	5008
Closing Stock	Closing Bal.Qty	8767	20784	2.370708	3.816747	-4	1	1	2	118
Closing Stock	Closing Bal.Val	8767	7667412.49	874.5765	1269.999	-756	399	604.2	898.51	40082.24

Table **MTT-08**: Descriptive statics for Opening and Closing Stock

Between April 1st 2023 and September 30th 2025, the average unit cost increased modestly from ₹450 to ₹461, while the average retail price rose from ₹688 to ₹710, indicating a gradual upward trend in product pricing. Inventory levels also grew, with total quantity rising from 13,984 to 20,784 units and average quantity per SKU increasing from 2.22 to 2.37. This expansion was accompanied by a jump in total inventory value from ₹52.27 lakh to ₹76.67 lakh, suggesting either increased stocking or a shift toward higher-value SKUs. However, the presence of zero or near-zero cost prices and negative quantities and values in the closing stock highlights potential data quality issues that warrant review.

Purchase and Purchase Returns: Descriptive statistics for Purchase and Purchase Returns are provided in the Table **MTT-09** below.

Dataset	Metric	Count	Sum	Mean	St.Dev.	Min	25%	50%	75%	Max
Purchase	Retail Price	48338	34451697	712.7249	395.1035	35	439	599	929	5008
Purchase	Purchase Qty	48338	138133	2.857648	5.576349	1	1	1	2	200
Purchase	Net Amount	48338	51228130.21	1059.79	1773.829	0.02	442.97	689.45	1049.303	71332.8
Purchase	Unit Price	48338	23884320	494.1106	268.2865	0.02	298.15	439.166	647.18	3144.76
Purchase Returns	Retail Price	10803	6816698	631.0005	360.2223	35	379	579	799	2499
Purchase Returns	Purch. Return Qty	10803	19937	1.845506	1.967594	1	1	1	2	25
Purchase Returns	Net Amount	10803	7618004.85	705.1749	669.8567	0.03	328.34	515.6	786.84	7694.45
Purchase Returns	Unit Price	10803	4756575	440.3013	246.2168	0.03	264.24	407.74	562.67	1749.3

Table **MTT-09**: Descriptive statics for Purchase and Purchase Returns

Purchased items had an average retail price of ₹713 and unit cost of ₹494, with wide variability across SKUs. Most purchases and returns were small in quantity, but a few bulk transactions skewed the totals. Returned items were generally lower in value, with an average retail price of ₹631 and unit cost of ₹440 when compared to purchased averages. This suggests that returns are concentrated in lower-priced SKUs, though a few high-value returns had notable financial impact.

Stock Verification: Descriptive statistics for Stock Verification are provided in the Table **MTT-10**.

Dataset	Metric	Count	Sum	Mean	St.Dev.	Min	25%	50%	75%	Max
Stock Verification	Retail Price	7034	5011520	712.4709	408.4211	35	399	613.5	949	3999
Stock Verification	Book Stock	7034	15259	2.16932	4.135061	-3	1	1	2	198
Stock Verification	Trans Qty. (Phy Stk Qty)	7034	15597	2.217373	3.354682	0	1	1	2	56
Stock Verification	Difference Qty	7034	338	0.048052	2.700716	-198	0	0	0	28
Stock Verification	Physical Stock Value	7034	5847014	831.2503	1041.239	0	367.83	602.95	902.42	18410

Table **MTT-10**: Descriptive statistics for Stock Verification

The Stock Verification dataset comprises 7,034 entries and reflects post-audit adjustments to the inventory. Retail prices range widely from ₹35 to ₹3,999, with a mean of ₹712.47, indicating a mix of low- and high-value SKUs. Book and physical stock quantities are closely aligned, though a small subset shows large corrections. These statistics confirm that while most items required minimal correction, a subset of SKUs experienced significant inventory mismatches.

Sales and Customer Returns: Descriptive statistics for Sales and Customer Returns are provided in the Table **MTT-11** below.

Dataset	Metric	Count	Sum	Mean	St.Dev.	Min	25%	50%	75%	Max
Sales	Sales Qty	91486	114534	1.251929	1.287232	1	1	1	1	100
Sales	Retail Price	91486	51147563	559.0753	344.6494	35	329	499	669	3999
Sales	Bill Level Disc. Amt.	91486	4677066.77	51.12331	157.2864	0	0	9	64.54	7328.19
Sales	Net Amount	91486	55070992.39	601.9609	451.4029	0.01	339	515.82	747.945	15571.81
Sales	Unit Price	91486	47563059.78	519.8944	315.592	0.01	305.1	466.935	629	2974.22
Sales	Effective Disc %	91486	N/A	6.348736	8.360742	0	0	2.912621	10.0377	99.99777

Returns	Sales Qty	2291	-2614	-1.14099	0.520103	-11	-1	-1	-1	-1
Returns	Retail Price	2291	1515984	661.7128	388.4977	35	399	579	849	3299
Returns	Bill Level Disc. Amt.	2291	-107451.63	-46.9016	84.16688	-949	-69.9	0	0	0
Returns	Net Amount	2291	-1526902	-666.478	375.7679	-2994	-852.05	-579	-409.18	-35
Returns	Unit Price	2291	1416401.808	618.2461	350.389	35	384.9	549	799	2705.18
Returns	Effective Disc %	2291	N/A	5.524026	8.035961	0	0	0	10	50

Table **MTT-11**: Descriptive statistics for Sales and Customer Returns

Sales transactions averaged 1.25 units per bill, with a mean retail price of ₹559 and an average unit price of ₹520 after discounts. Most bills had minimal discounting, but a few high-value transactions pushed the effective discount rate up to nearly 100%, indicating potential data anomalies or promotional overrides. Customer returns were concentrated in lower-quantity, higher-priced SKUs, with an average unit price of ₹618, higher than the sales average, suggesting that premium products may be more prone to returns. The financial impact of returns is notable, with over ₹15 lakh reversed from net revenue.

Expenses: Descriptive statistics for expenses are provided in the Table **MTT-12** below.

Dataset	Metric	Count	Sum	Mean	St.Dev.	Min	25%	50%	75%	Max
Monthly Expense	expense	30	10644896	354829.9	70477.75	234748	309847.5	344202.5	395080.5	577102

Table **MTT-12**: Descriptive statistics for Expense data

Monthly operating expenses averaged ₹3.55 lakh over the 30-month period, with moderate variability (Std. Dev. ₹70,478). The lowest recorded expense was ₹2.35 lakh, while the highest reached ₹5.77 lakh, suggesting occasional spikes likely tied to seasonal costs or one-off operational events. Overall, expenses remained relatively stable, with 50% of months falling between ₹3.09 lakh and ₹3.95 lakh.

4. Analysis Process, Tools and Methods

The following section describes the process flow for the project, tools developed and the methods used to achieve the stated goals.

4.1 Process Flow

The core work for the project is divided into three parts: Pre-processing, processing and the Conclusion stages. Following diagram **MTF-01** shows the activities in each stage.

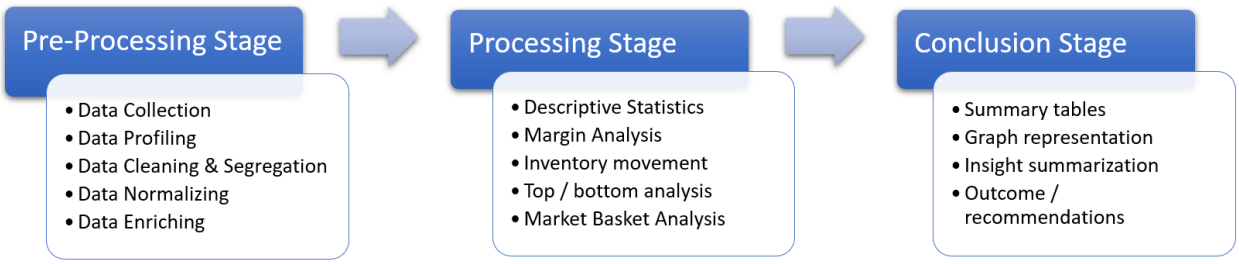

Figure **MTF-01**: Process Stages and their breakdown

Data Collection: Multiple datasets have been collected from the POS system in CSV format, for dates between April 1st 2023 and September 30th 2025. In addition, collected expense information from the Tally system and manually created the expense data file.

Data Profiling: Performed initial scans on the collected data for both categorical and numerical values to assess the volume, column types, and missing values. Since categorical data is incomplete and inconsistent, primarily relied on quantitative data for the analysis. In addition, noticed the sales and sale voucher numbers are recycled for each year. For example, S/1598 records are repeated for 2023, 2024 and 2025. The solution is to create a unique key by combing voucher number and date.

Data Cleaning and Segregation: With the goal to **not** modify the collected data, data cleaning and data segregation is done programmatically. For example, removed summary lines like grand total using *dataframe.drop()* method. Used conditional copy for segregation, for example *sales_df = df[df['Sales Qty'] > 0].copy()* and *returns_df = df[df['Sales Qty'] < 0].copy()* since Sales and Customer Return data are mixed into one dataset.

Data Normalizing: Identified and standardized numeric data columns for consistency and computation using *dataframe.apply()* method.

Data Enriching: Enhanced the datasets by computing diagnostic columns such as Retail Value, Unit Price, and Effective Discount %, for example *df['Unit Price'] = df['Net Amount'] / df['Sales Qty']*

Processing Stage: This stage focuses on detailed analysis aligned with specific business outcomes. Descriptive statistics have been covered in earlier sections. Additional diagnostics, such as margin analysis, inventory movement, and return impact are discussed in subsequent sections.

Conclusion Stage: The conclusion stage translates processed data into structured business outputs. *Summary tables* are used to present computed metrics with consistent column order, ensuring clarity and comparability across datasets. *Graphical representations* are selectively used to highlight trends, anomalies, or outliers that warrant attention. *Insight summarization* uses key findings from descriptive statistics into concise, actionable interpretations. *Final outcomes and recommendations*, such as pricing inconsistencies, high-return SKUs, or supplier-level concerns, are addressed in subsequent sections.

4.2 Tools

This project is being built entirely in **Python**, selected for its flexibility, rich ecosystem, and precise control over data workflows. The goal is to build a plug-and-play architecture system supporting growing collection of analyzer tools. **R** and **Google Colab** were initially considered but ruled out due to limitations in flexibility, cloud dependency, and lack of packaging support for long-term business

deployment. Figure **MTF-02** shows code snippet of a generic method to compute descriptive statistics and a sample method call.

```python
def __get_descriptive_statistics(self, df, dfname, selected_columns):
    stats_df = df[selected_columns].describe().transpose()
    stats_df.index.name = 'metric'
    stats_df['dataset'] = dfname
    sum_values = df[selected_columns].sum(numeric_only=True)
    stats_df['sum'] = sum_values
    stats_df = stats_df.reset_index()
    stats_df = stats_df[['dataset', 'metric', 'count', 'sum'] +
                        [col for col in stats_df.columns if col not in ['dataset', 'metric', 'count', 'sum']]]
    return stats_df

    # description stats for purchase return data
    df = self.app_container.dataset.get_data(DataTypes.PURCHASE_RETURN_DATA)
    selected_columns = df.select_dtypes(include='number').columns.tolist()
    stats = self.__get_descriptive_statistics(df, "Purchase Returns", selected_columns)
    combined_stats = pd.concat([combined_stats, stats])
```

Figure **MTF-02**: Python code snippet to compute Descriptive Statistics and Sample caller

The code is still evolving and is being constantly refactored, so complete architecture will be documented in the Final Report. A partial version of code is available on the google shared drive.

4.3 Analytical Methods by Domain

This project adopts a modular, diagnostic-first approach to retail analytics, with each method aligned to a specific business objective. The emphasis is on interpretability, reproducibility, and business relevance with transparent logic and structured outputs.

Exploratory Data Analysis: ***EDA*** was conducted during initial profiling to assess data structure, volume, and integrity. Summary statistics and distribution plots were used to identify outliers, pricing anomalies, and return behavior, informing the design of downstream diagnostics.

Margin Diagnostics: This is implemented through two complementary methods. First, ***Realized and Unrealized Margins*** were computed using aggregated sales, purchase, and return data. This enabled identification of top and bottom margin SKUs and pricing inconsistencies. Second, ***Margin Change Analysis*** compared opening and closing stock to quantify margin *erosion* or *gain*, using weighted impact based on closing quantity. Distribution plots of retail value and unit price further revealed asymmetries between sales and customer returns.

Inventory Movement Classification: Sales and stock data are used to estimate *SKU Velocity*. Items will be categorized into fast-moving, slow-moving, and deadstock based on movement frequency and closing quantity. Thresholds are being refined to reflect business context and seasonal variation.

Sales Basket Modeling: This is scheduled for the next phase. It will apply the *Apriori algorithm* to transactional data to identify frequently co-purchased items, supporting bundling strategies and cross-selling opportunities.

Operational Insight: This is a documentation exercise for process mapping, highlighting traceability challenges in batch-level pricing and SKU identification. These limitations will inform future system design and data capture improvements.

5. Results and Findings

Complete raw data and graphs for Margin Diagnostics and Inventory Analysis are available in *result_margin.xlsx*, *result_inventory.xlsx* files in the output directory of the project folder.

5.1 Margin Diagnostics

Top performers for metrics - Realized Margin, Unrealized Margin, Margin Loss, Sale Amount, Sale Quantity, Margin Erosion and Margin Gain are all computed and are produced in *result_margin.xlsx* file. Figure MTF-03 shows Top 5 for Realized Margin, Sales Quantity and Sales Amount (Full size pictures are uploaded to images directory).

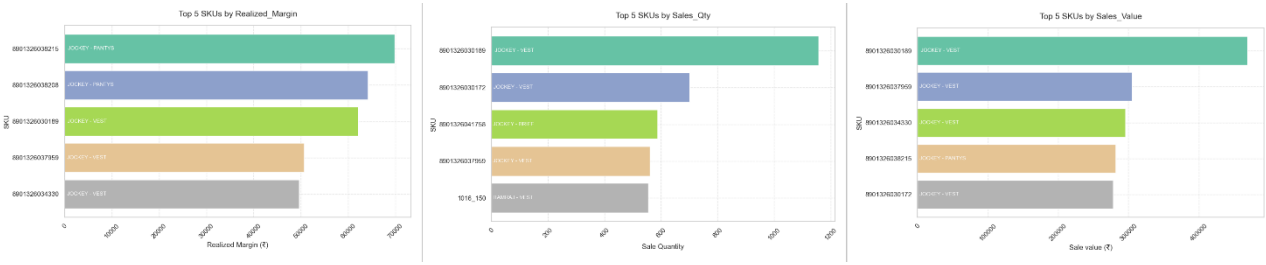

Figure **MTF-03**: Top 5 Realized Margin, Sales Quantity and Sales Amount

Looking across margin, sale quantity and value, a few SKUs really stand out. The top seller by both quantity and value is 8901326030189 (JOCKEY - VEST), but interestingly, it doesn't show up in the top margin list, which probably means it's priced too low. On the flip side, 8901326038125 (JOCKEY - PANTYS) has the highest realized margin but barely sells, so there's likely untapped potential there. Then there's 8901326037959 (JOCKEY - VEST), which performs well across all three metrics, a solid all-rounder worth protecting and scaling. Overall, there's a clear opportunity to rethink how Eesbee can promote and price these SKUs: push the high-margin ones harder, dig into cost structures for the high-volume ones, and double down on the balanced performers.

Margin Gain and Erosion: Across the assortment, margin dynamics were shaped by shifts in closing stock valuations. A gain in margin was realized across **452 SKUs**, contributing a cumulative uplift of **₹116,407**. In contrast, **65 SKUs** experienced margin erosion, resulting in a total negative impact of **₹7,284**. While erosion was limited in scope, the broader trend reflected a net improvement in profitability, with over **30%** of SKUs enhancing their margin position.

Unit Price Distribution: The *KDE (Kernel Density Estimate) plot* shown in Figure **MTF-04** is comparing unit price distributions for sales and returns, and it reveals distinct behavioral patterns across transactions. Sales are concentrated in lower price bands, with a smooth density peak suggesting consistent movement of affordable items. In contrast, the return distribution shows a longer tail and higher density in premium price segments, indicating that higher priced products are more prone to being returned. This divergence highlights potential mismatches in customer expectations, product quality, or value perception. Eesbee should review its pricing strategy, product positioning, or return policy refinement to limit returns.

Figure **MTF-04**: Unit Price Distribution

5.2 Inventory Analysis

Inventory analysis is work in progress, hence graphs for some of these will be produced for the *Final Report*. The computed data for inventory analysis is available in *result_inventory.xlsx* file.

Negative Stock: Few SKUs are showing negative closing balances across brands like VIP, RAMRAJ, VSTAR etc. These findings need validation against physical inventory and transaction history.

Stocked but Unsold Returns Report: A total of **2,230 SKUs** was stocked but show *no sales and no closing inventory*, indicating they were likely returned after shelving. These items ***occupied shelf space and operational effort*** without contributing to revenue, with a cumulative return value of **₹14.28 lacks**. Sharing this insight with business to support inventory efficiency improvements, targeting these SKUs can help reduce non-converting stock and optimize assortment planning.

Dead Stock Report: The deadstock report found **3,928 SKUs** with a total purchase value of **₹28 lakhs** and unrealized margin of **₹14.47 lakhs**, reflecting significant capital tied up in non-moving inventory. Many of these items show recent purchases but zero sales, indicating poor sell-through and potential overstocking. These SKUs should be prioritized for markdowns, liquidation, or repurposing to recover value and improve inventory efficiency.

Discrepancy Stock Report: The discrepancy stock report identified **41 SKUs** where actual closing quantities deviate from expected values, despite clear purchase and sales activity. These gaps may stem from mis-scans, unrecorded returns, or verification errors, and should be prioritized for physical audit and system correction to maintain inventory integrity.